LECTURE 3

Def'n Let $\{a_n\}$ be a sequence. We call $\sum_{n=1}^{\infty} a_n$ a series

Ex The sequence $\left\{\frac{1}{n}\right\}$ gives the series $\sum_{n=1}^{\infty} \frac{1}{n}$

Def'n $\left\{\frac{1}{n}\right\}$ is called the <u>harmonic series</u>.

Ex Find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{2^n}$.

$$a_1 = \frac{1}{2}$$

$$a_1 + a_2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$a_1 + a_2 + a_3 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$$

$$a_1 + a_2 + a_3 + a_4 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{15}{16}$$

We have $S_k = \frac{2^k - 1}{2^k}$. Then,

$$\begin{aligned}
\lim_{k \to \infty} S_k &= \lim_{k \to \infty} \frac{2^k - 1}{2^k} \\
&= \lim_{k \to \infty} 1 - \frac{1}{2^k} \\
&= 1 - 0 \\
&= 1
\end{aligned}$$

Hence $\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$.

Def'n Let $\{a_n\}$ be a sequence and $\sum_{n=1}^{\infty} a_n$ be a series.

We define S_k by $S_k = \sum_{n=1}^{k} a_n$ the <u>k^{th} partial sum</u>.

series, harmonic series, k^{th} partial sum

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Def'n The new sequence $\{s_k\}$ is the <u>sequence of partial sums</u>.

If $\lim_{k \to \infty} S_k = S$, then $\sum_{n=1}^{\infty} a_n$ <u>converges</u> to S and $\sum_{n=1}^{\infty} a_n = S$.

If the limit DNE, the series <u>diverges</u>.

Theorem <u>Divergence Test</u>

If $\lim_{n \to \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Ex Show $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)$ diverges.

We have $\lim_{n \to \infty} 1 + \frac{1}{n} = 1 + 0 = 1 \neq 0$.

Hence the series diverges by the Divergence Test

Remark (1) The Divergence Test can only be used to prove divergence.

If $\lim_{n \to \infty} a_n = 0$ then the test does not hold.

Ex $\lim_{n \to \infty} \frac{1}{n} = 0$ but the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

(2) The Divergence Test can be used to prove the limit of a sequence is 0 using the contrapositive.

Def'n $\left\{ \frac{(-1)^{n+1}}{n} \right\}$ is called the <u>alternating harmonic series</u>.

Ex $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = \ln(2)$. Hence it converges.

sequence of partial sums, divergence test, alternating harmonic series